

Robert Yav

MS Statistics Candidate · San José State University

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EDUCATION

MS Statistics (In Progress)

Expected May 2028

San José State University, San José, CA

Coursework: Stochastic Processes · Design and Analysis of Experiments · Regression Theory · Mathematical Data Visualization

BS Applied Mathematics

Graduated May 2023

Concentration: Statistics · Minor: Computer Science · San José State University · GPA: 3.62 (Cum Laude)

Coursework: Probability & Statistics I–II · Mathematical Statistics · Numerical Analysis & Scientific Computing · Linear and Nonlinear Optimization · Mathematical Modeling · Data Structures & Algorithms · Fundamentals of Data Science · Programming in R, Python & SQL · Object-Oriented Programming · Combinatorics

SKILLS

Statistical Methods: OLS & LASSO Regression, ANOVA, Friedman Test, Cross-validation (k-fold), Stochastic Processes, Experimental Design

Machine Learning: Random Forest, SVM, XGBoost, Naive Bayes, KNN, Logistic Regression, Gradient Boosting, NLP (spaCy)

Dimensionality Reduction: PCA, t-SNE, UMAP · SMOTE & Image Augmentation for class imbalance

Languages: Python (NumPy, Pandas, scikit-learn, matplotlib), R (tidyverse, ggpubr, caret), SQL, Java, LaTeX

Tools: JMP (SAS), Quarto, Excel, Git/GitHub

PROJECTS

Human vs AI Art Perception — Replication Analysis

Spring 2026

github.com/meganajoseph/AI_sentiments

- Reshaped a 60-participant × 319-column survey dataset from wide to long format in R tidyverse for analysis.
- Audited and reanalyzed the original study's Friedman Test and Within-Factors Repeated Measures ANOVA, uncovering dataset mislabeling through systematic comparison of descriptive statistics using ggpubr.

Traffic Sign Classification

Spring 2026

github.com/rubertyao/NCAAWomensVolleyballAnalysis

- Built a computer vision classification pipeline on the LISA Traffic Sign dataset, handling significant class imbalance via image augmentation and SMOTE.
- Compared PCA, t-SNE, and UMAP dimensionality reduction across Random Forest, SVM, Naive Bayes, and XGBoost classifiers to identify optimal combinations.

Predicting BMI from Screentime in Chinese Adolescents

Fall 2025

github.com/rubertyao/ScreentimeBMI

- Applied OLS regression, LASSO, and variable transformations to predict adolescent BMI from screentime and demographic predictors using a Chinese national health dataset.
- Compared model performance via k-fold cross-validation using RMSE, R^2 , and adjusted RMSE.

Natural Disaster Tweet Classification

2023

ML Club @ SJSU (Team Lead)

- Led a team to classify social media posts as describing real vs. non-real natural disasters using NLP preprocessing (spaCy lemmatization and tokenization).
- Benchmarked Multinomial Naive Bayes, Logistic Regression, KNN, Decision Trees, Random Forest, SVM, and Gradient Boosting across preprocessing configurations.

EXPERIENCE

Math Lab Lead · Instructional Support Assistant

2024 – Present

Evergreen Valley College, San José, CA

- Lead the Math, Science & Engineering Resource Center (MSRC), extending lab hours into evening sessions to expand student access.
- Collaborate with faculty, staff, and student tutors to maintain educational quality and support student confidence across STEM disciplines.

Instructional Student Assistant

2022 – 2023

San José State University, San José, CA

- Facilitated workshops for MATH 70X (Mathematics for Business) and MATH 30X (Calculus I), improving course consistency across sections under supervising professors.